Raw Data Profiling Summary

Online Retail II Data Pipeline Project

Overview

Before building the staging layer and the analytics model, I first profiled the raw dataset to understand its structure, identify potential data quality issues, and determine which cleaning rules would be necessary.

The dataset contains 1,067,371 rows and is structurally complete. Key fields such as invoice number, stock code, quantity, and price are present for all records.

The goal of this profiling step was to identify patterns in the raw data and determine which records should be retained, cleaned, or excluded when building the staging layer.

Dataset Description

The project uses the Online Retail II dataset, which contains transactional records from a UK-based online retail company. The dataset covers sales activity between December 2009 and December 2011.

The data is provided in two CSV files:

- Year_2009-2010_online_retail_II.csv
- Year_2010-2011_online_retail_II.csv

Each row represents a single product within an invoice. Because invoices may contain multiple products, several rows can share the same invoice number.

The dataset includes the following fields:

- Invoice — unique invoice identifier. Invoices beginning with the letter *"C"* represent cancelled transactions.
- StockCode — product identifier used by the retailer.
- Description — text description of the product.
- Quantity — number of units purchased. Negative values represent returned items.
- InvoiceDate — date and time when the transaction occurred.
- Price — price per unit of the product.
- Customer ID — identifier assigned to the customer. Some transactions do not include a customer identifier.
- Country — the country where the customer is located.

Across the two files, the dataset contains 1,067,371 rows, representing more than one million product-level transactions.

Description Field Quality

During profiling, I observed that 213,035 rows contained leading or trailing spaces in the product description field.

Although this does not affect numerical calculations, it can create problems when grouping or comparing product descriptions.

To address this issue, the staging process applies: TRIM(description)

This ensures that product descriptions are consistent and avoids duplicate labels caused by extra whitespace.

Customer Information

The dataset contains 243,007 rows where customer_id is NULL.
After examining these records, it became clear that these do not represent missing data errors.
Instead, they correspond to transactions where the customer was not identified.
Because these transactions still represent valid sales activity, the records were retained in the dataset.

Returns and Cancellations

Returns and cancellations are clearly identifiable in the dataset.
Two patterns appear consistently:

- 22,950 rows contain negative quantities, which represent returned items.
- 19,494 invoices begin with the letter "C", indicating cancellation transactions.

These records represent legitimate business activity rather than data errors. As a result, they were retained in the dataset to analyze return behavior and accurately calculate net revenue.

Pricing Issues

A small number of pricing anomalies were identified during profiling.

- 5 rows contained negative prices, which appear to be data entry errors. These records were excluded during staging.
- 6,220 rows contain a price of zero. These likely represent promotional items, free samples, or bundled products. Because these transactions can still be part of real sales activity, they were retained.

Extreme Values

Some transactions contain unusually large quantities or prices. After reviewing these records, they appear to reflect bulk purchases or special adjustments rather than data corruption.
Since these values are consistent with possible business scenarios, they were retained.

Non-Product Transactions

A small set of records appears to represent operational or financial adjustments rather than merchandise sales.
Examples include:

- stockcode = 'M'
- descriptions such as BANK CHARGES or AMAZONFEE

Because these transactions do not represent actual product sales, they were excluded from the cleaned staging dataset.

Outcome of the Profiling Step

Profiling the raw dataset allowed me to design the staging transformations intentionally rather than applying arbitrary cleaning rules. Each transformation implemented in the staging layer was derived directly from patterns observed in the raw data during the profiling process.
By documenting these decisions, the pipeline remains transparent and reproducible, and each transformation can be traced back to a specific observation in the raw dataset. Once these rules were implemented, the dataset became suitable for building the analytics model and performing downstream analysis.